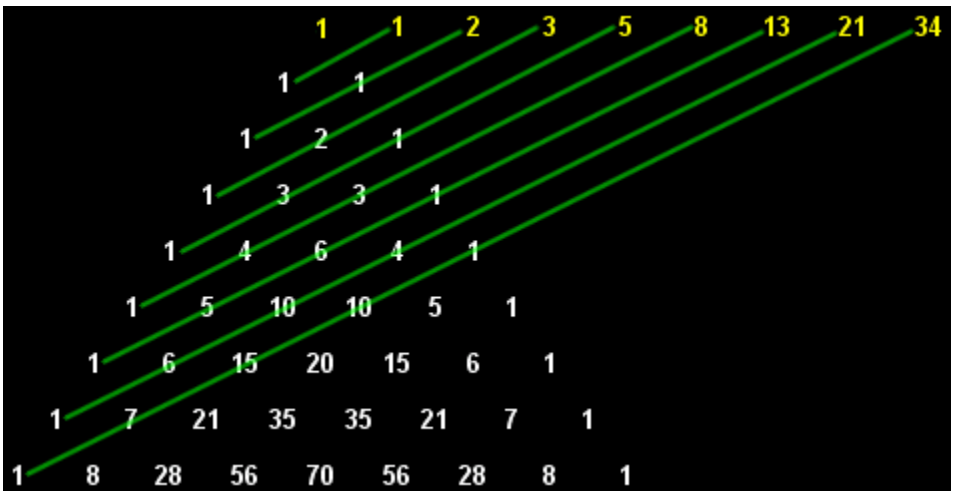
American Computer Science League

2021-2022 ● Contest 3: Fibonacci & Pascal ● Intermediate Division

PROBLEM: One of the many patterns that can be found in Pascal's Triangle is where the Fibonacci sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, … can be found by adding the numbers on the diagonals as illustrated with the following diagram:



In this program, generate each row of Pascal's Triangle by placing a 1 in the first and last position. For the positions in the middle of the triangle, every number is the sum of the numbers immediately above it. The 0th row contains a single 1. For row N, there are N+1 items.

INPUT: There are 5 lines of data. Each line has a single integer representing one of the Fibonacci numbers in the following sequence: 1 1 2 3 5 8 13 21 34 55 89 .... We guarantee it will be less than 2^{31} since that is the largest integer that can be represented using 32 bits.

OUTPUT: For each line of data, find the integers on the diagonal of Pascal's Triangle that add up to the given number. Then print the number of odd values, the number of even values, and the largest value all separated by a single space.

SAMPLE INPUT:

```
8
89
610
10946
317811
```

SAMPLE OUTPUT:

```
1. 2 1 4
2. 5 1 35
3. 4 4 210
4. 8 3 3003
5. 3 11 77520
```

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TEST DATA

TEST INPUT:

```
55
1597
832040
9227465
1836311903
```

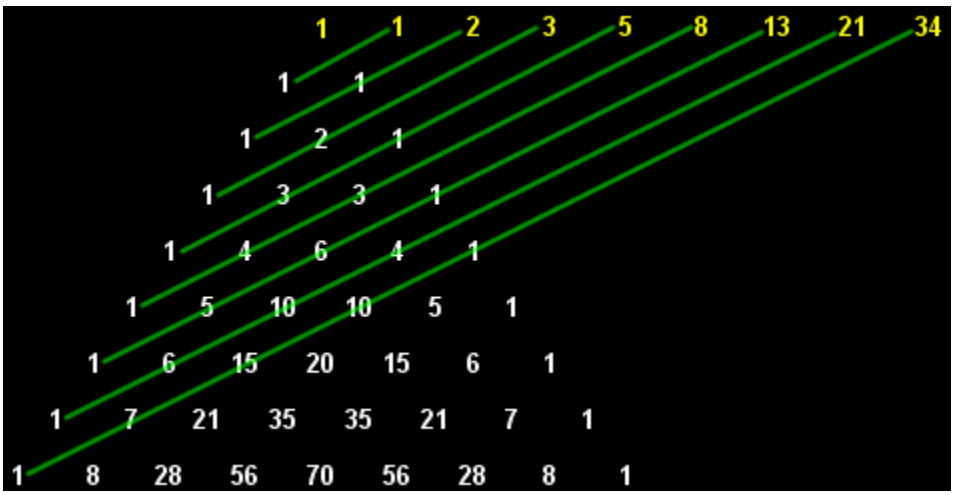
TEST OUTPUT:

```
1. 3 2 21
2. 5 4 495
3. 4 11 203490
4. 9 9 2042975
5. 7 16 354817320
```

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题目：在帕斯卡三角形中能够发现一些性质，如通过将**帕斯卡三角形**对角线上的数字相加，**便可以找到**斐波那契数列 1, 1, 2, 3, 5, 8, 13, 21, 34, …, **如下图所示：**



在上图中，帕斯卡三角形的每一行的首尾位置都是 1 。**在**三角形中间位置**上**的每个数字都是其上方紧邻的两个数字之 。第 0 行只有一个数字 1 。 第 N 行有 N+1 个数字。

输入：输入 5 行数据。每行**都**有一个整数，表示斐波那契数列中的**某个元素**： 1 1 2 3 5 8 13 21 34 55 89 .... 我们保证所有数字都小于 2^{31} ，因为 2^{31} 是可以用 32 位表示的最大整数。

输出: 根据输入的每行数据，找出帕斯卡三角形中某条对角线上的所有整数，**这些**整数**相加之和等于给定整数。并**输出奇数的个数、偶数的个数和最大值，中间由一个空格隔开。

示例输入:

```
8
89
610
10946
317811
```

示例输出:

```
1. 2 1 4
2. 5 1 35
3. 4 4 210
4. 8 3 3003
5. 3 11 77520
```

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测试数据

测试输入：

```
55
1597
832040
9227465
1836311903
```

测试输出：

```
1. 3 2 21
2. 5 4 495
3. 4 11 203490
4. 9 9 2042975
5. 7 16 354817320
```